

State Transition Testing

A technique for validating behavior when outcomes depend on the current state and occurring events.

Model

Identify states, events, valid transitions, invalid transitions and resulting actions.

Examples

Order: created → paid → shipped → delivered; account: active → blocked → reactivated; session: authenticated → expired.

Coverage

Test valid paths, forbidden events, repetition, incorrect order and final states.

History

Defects may depend on the previous sequence; record initial state and events to reproduce correctly.

When to use

Use this knowledge when risk, architecture or the development lifecycle stage requires specific validation. Before applying it, confirm the objective, environment, required data and expected evidence.

How QA applies it in practice

Start from the behavior or risk that needs validation, prepare data and environment, execute the technique or tool reproducibly and record evidence that allows the team to understand the result.